Case Study: COVID-19

COVID-19 Case Study:

Overview: *The COVID-19 pandemic, caused by the SARS-CoV-2 virus, emerged in late 2019 and rapidly spread globally, leading to significant health, economic, and social impacts. This unprecedented health crisis highlighted the crucial role of data analysis in managing such pandemics. By meticulously tracking and analyzing data on confirmed cases, recoveries, and deaths, policymakers and health professionals can make informed decisions to control the spread of the virus and allocate resources effectively.*

Dataset Details:
Dataset *(Open link in a new tab and download the dataset.)*
This case study utilizes three key datasets, each providing daily updates on different aspects of the pandemic for various countries and regions:

- **Confirmed Cases Dataset:** *Contains the cumulative number of confirmed COVID-19 cases per day for each country and region. The data spans from January 22, 2020, to May 29, 2021, with over 276 geographic entries.*
- **Deaths Dataset:** *Records the cumulative number of deaths attributed to COVID-19, structured similarly to the confirmed cases dataset. This provides crucial information for assessing the lethality and outbreak severity in different areas.*
- **Recovered Cases Dataset:** *Includes data on the cumulative number of individuals who have recovered from COVID-19, which is vital for understanding the disease's progression and the effectiveness of treatment protocols.*

Each dataset includes columns for Province/State, Country/Region, geographic coordinates (Lat, Long), and a series of dates representing daily cumulative totals.

Objective of the Case Study:
The primary objectives of this case study are:

- **Practical Application of Python:** *To demonstrate and enhance practical skills in using Python for data analysis, focusing on data manipulation, cleaning, and visualization.*
- **Insightful Data Analysis:** *To provide analytical insights into the dynamics of COVID-19's spread, recovery, and mortality rates across different regions and over time.*
- **Skill Development:** *To equip students with the knowledge to handle real-world data using Python libraries such as Pandas for data manipulation, Matplotlib for data visualization, and Numpy for numerical data processing.*

Analysis Questions:

1. ***Data Loading:*** *How do you load the COVID-19 datasets for confirmed cases, deaths, and recoveries into Python using Pandas?*

2. ***Data Exploration***

   1. *After loading the datasets, what is the structure of each dataset in terms of rows, columns, and data types?* `.info()`
   2. *Generate plots of confirmed cases over time for top countries.*
   3. *Generate plots of confirmed cases over time for China.*

3. ***Handling Missing Data:*** *Identify these missing values and replace them using a suitable imputation method such as forward filling for time-series data.*

4. ***Data Cleaning and Preparation:*** *Replace blank values in province column with "All Provinces"*

5. ***Independent Dataset Analysis***

   1. *Analyze the peak number of daily new cases in Germany, France, and Italy. Which country experienced the highest single-day surge, and when did it occur?*
   2. *Compare the recovery rates (recoveries/confirmed cases) between Canada and Australia as of December 31, 2020. Which country showed better management of the pandemic according to this metric?*
   3. *What is the distribution of death rates (deaths/confirmed cases) among provinces in Canada? Identify the province with the highest and lowest death rate as of the latest data point.*

6. ***Data Transformation***

   1. *Transform the 'deaths' dataset from wide format (where each column represents a date) to long format where each row represents a single date and columns are now country names, ensuring that the date column is in datetime format. How would this transformation be executed?* `.melt()`
   2. *What is the total number of deaths reported per country up to the current date?*
   3. *What are the top 5 countries with the highest average daily deaths?*
   4. *How have the total deaths evolved over time in the United States?*

7. *Data Merging*

1. *How would you merge the transformed datasets of confirmed cases, deaths, and recoveries on the 'Country/Region' and 'Date' columns to create a comprehensive view of the pandemic's impact?* `Merged Table` ✅
2. *Analyze the monthly sum of confirmed cases, deaths, and recoveries for countries to understand the progression of the pandemic.[From the merged dataset]* ✅
3. *Redo the analysis in last question for United States, Italy and Brazil.* ✅

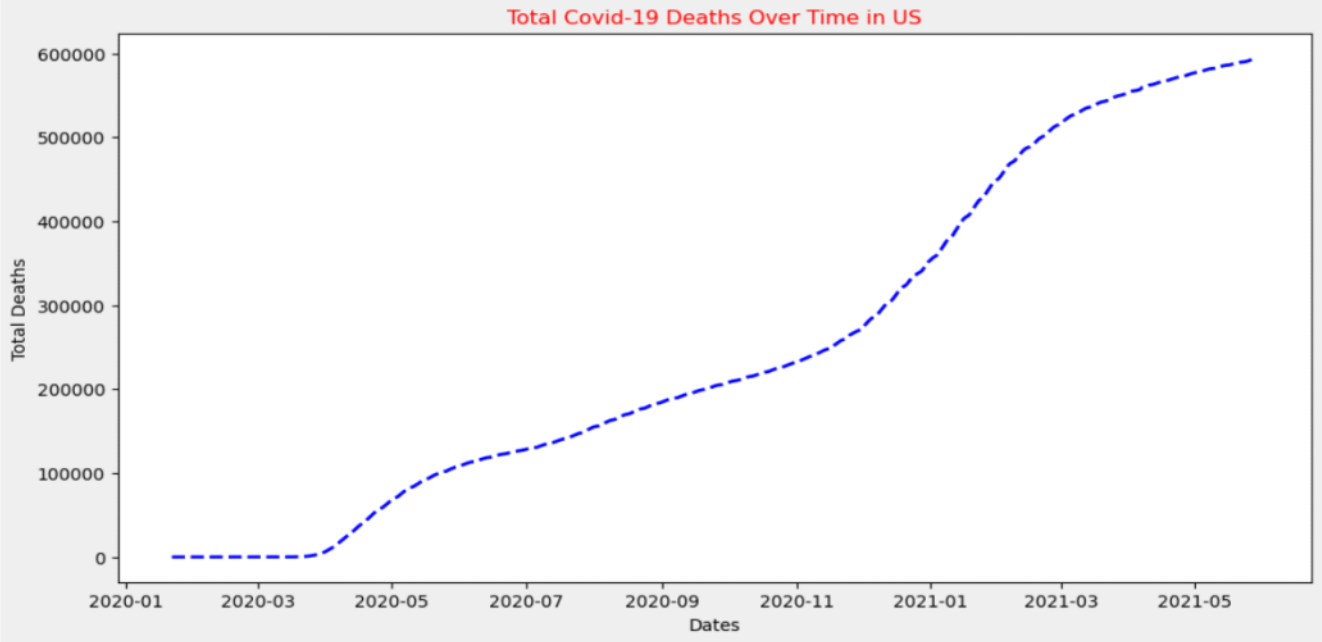
8. *Combined Data Analysis*

1. *For the combined dataset, identify the three countries with the highest average death rates (deaths/confirmed cases) throughout 2020. What might this indicate about the pandemic's impact in these countries?* ✅
2. *Using the merged dataset, compare the total number of recoveries to the total number of deaths in South Africa. What can this tell us about the outcomes of COVID-19 cases in the country?* ✅
3. *Analyze the ratio of recoveries to confirmed cases for the United States on a monthly basis from March 2020 to May 2021. Which month experienced the highest recovery ratio, and what could be the potential reasons?* ✅

```python
# 6.4 How have the total deaths evolved over time in the United States? ✗✗
us_deaths_over_time = (
    deaths_long[deaths_long['Country/Region'] == 'US']
    .groupby('Date')['Deaths'].sum()
)
us_deaths_over_time
```

```
Date
2020-01-22         0
2020-01-23         0
2020-01-24         0
2020-01-25         0
2020-01-26         0
                ...
2021-05-25    590941
2021-05-26    591950
2021-05-27    593288
2021-05-28    593963
2021-05-29    594306
Name: Deaths, Length: 494, dtype: int64
```

```python
# Visualization [Line Plot] [Time Related Trend Axis]
plt.figure(figsize = (12,6) , facecolor = '#E8E8E8')
plt.plot(us_deaths_over_time.index , us_deaths_over_time.values, color = 'blue' , linewidth =2,
        linestyle = '--')
plt.title('Total Covid-19 Deaths Over Time in US' , color = 'red')
plt.xlabel('Dates')
plt.ylabel('Total Deaths')
plt.show()
```



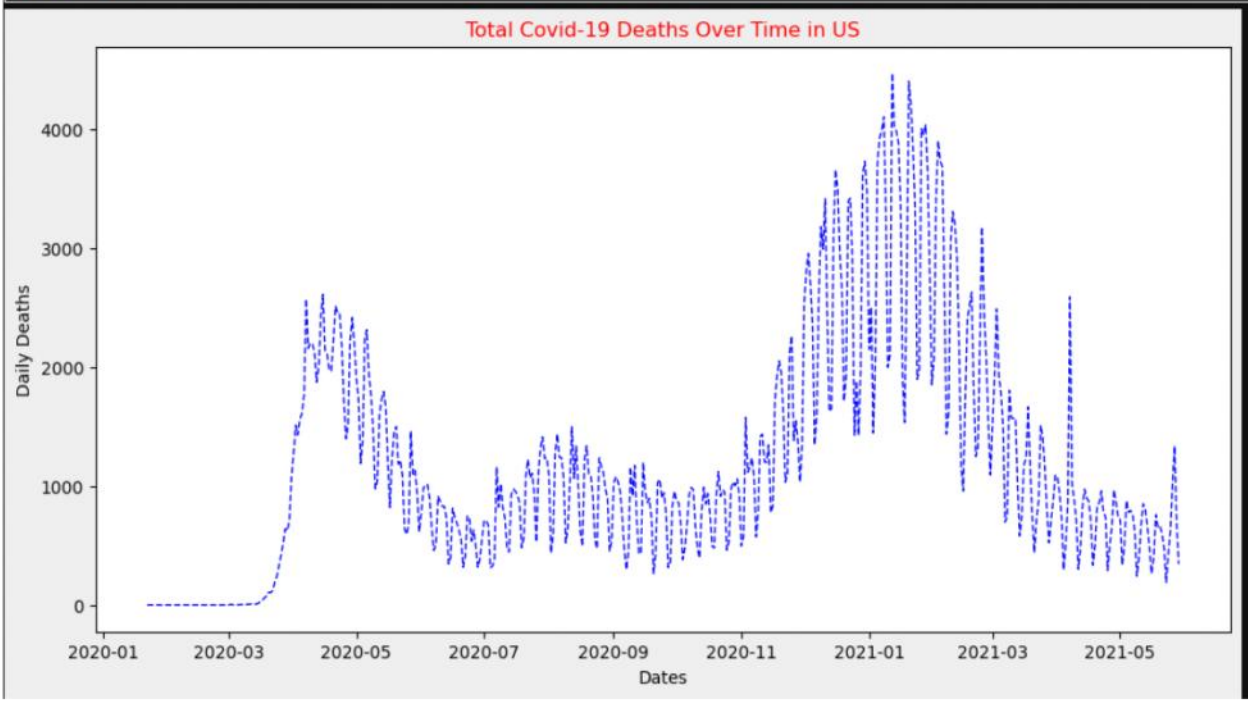

```python
us_deaths_over_time = (
    deaths_long[deaths_long['Country/Region'] == 'US']
    .groupby('Date')['Deaths'].sum()
    .reset_index()
)
us_deaths_over_time.head()
```

	Date	Deaths
0	2020-01-22	0
1	2020-01-23	0
2	2020-01-24	0
3	2020-01-25	0
4	2020-01-26	0

```python
us_deaths_over_time['DailyDeaths'] = us_deaths_over_time['Deaths'].diff().fillna(0)
us_deaths_over_time
```

	Date	Deaths	DailyDeaths
0	2020-01-22	0	0.0
1	2020-01-23	0	0.0
2	2020-01-24	0	0.0
3	2020-01-25	0	0.0
4	2020-01-26	0	0.0
...	...	...	...
489	2021-05-25	590941	621.0

```python
# Visualization [Line Plot] [Time Related Trend Axis]
plt.figure(figsize = (12,6) , facecolor = '#E8E8E8')
plt.plot(us_deaths_over_time['Date'] , us_deaths_over_time['DailyDeaths'], color = 'blue' , linewidth =1,
        linestyle = '--')
plt.title('Total Covid-19 Deaths Over Time in US' , color = 'red')
plt.xlabel('Dates')
plt.ylabel('Daily Deaths')
plt.show()
```




```python
# 7.2 Analyze the monthly sum of confirmed cases, deaths, and recoveries for countries to
# understand the progression of the pandemic. [From the merged dataset]
m = merged.copy()
m = m.sort_values(['Country/Region' , 'Date'])
m['YearMonth'] = m['Date'].dt.to_period('M')
m
```

	Country/Region	Date	Confirmed	Deaths	Recovered	YearMonth
0	Afghanistan	2020-01-22	0	0	0	2020-01
1	Afghanistan	2020-01-23	0	0	0	2020-01
2	Afghanistan	2020-01-24	0	0	0	2020-01
3	Afghanistan	2020-01-25	0	0	0	2020-01
4	Afghanistan	2020-01-26	0	0	0	2020-01
...	...	...	...	...	...	...
95337	Zimbabwe	2021-05-25	38706	1587	36517	2021-05
95338	Zimbabwe	2021-05-26	38819	1589	36531	2021-05
95339	Zimbabwe	2021-05-27	38854	1592	36541	2021-05
95340	Zimbabwe	2021-05-28	38918	1592	36563	2021-05
95341	Zimbabwe	2021-05-29	38933	1594	36578	2021-05

95342 rows × 6 columns

```python
# Daily Confirmed
confirmed_diff = m.groupby('Country/Region')['Confirmed'].diff().clip(lower = 0)
m['DailyConfirmed'] = confirmed_diff.fillna(0)

# Daily Deaths
deaths_diff = m.groupby('Country/Region')['Deaths'].diff().clip(lower = 0)
m['DailyDeaths'] = deaths_diff.fillna(0)

# Daily Recovered
recovered_diff = m.groupby('Country/Region')['Recovered'].diff().clip(lower = 0)
m['DailyRecovered'] = recovered_diff.fillna(0)

m
```

	Country/Region	Date	Confirmed	Deaths	Recovered	YearMonth	DailyConfirmed	DailyDeaths	DailyRecovered
0	Afghanistan	2020-01-22	0	0	0	2020-01	0.0	0.0	0.0
1	Afghanistan	2020-01-23	0	0	0	2020-01	0.0	0.0	0.0
2	Afghanistan	2020-01-24	0	0	0	2020-01	0.0	0.0	0.0
3	Afghanistan	2020-01-25	0	0	0	2020-01	0.0	0.0	0.0
4	Afghanistan	2020-01-26	0	0	0	2020-01	0.0	0.0	0.0
...	...	...	...	...	...	...	...	...	...
95337	Zimbabwe	2021-05-25	38706	1587	36517	2021-05	10.0	1.0	46.0

```python
m.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 95342 entries, 0 to 95341
Data columns (total 9 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   Country/Region  95342 non-null  object
 1   Date            95342 non-null  datetime64[ns]
 2   Confirmed       95342 non-null  int64
 3   Deaths          95342 non-null  int64
 4   Recovered       95342 non-null  int64
 5   YearMonth       95342 non-null  period[M]
 6   DailyConfirmed  95342 non-null  float64
 7   DailyDeaths     95342 non-null  float64
 8   DailyRecovered  95342 non-null  float64
dtypes: datetime64[ns](1), float64(3), int64(3), object(1), period[M](1)
memory usage: 6.5+ MB
```

```python
monthly_sum_trends = (
    m.groupby(['Country/Region' , 'YearMonth'])[['DailyConfirmed','DailyDeaths','DailyRecovered']]
    .sum()
    .rename(
        columns = {
            'DailyConfirmed' : 'Confirmed',
            'DailyDeaths' : 'Deaths',
            'DailyRecovered' : 'Recovered',
        }
    )
    .astype('int64')
    .reset_index()
    .sort_values(['Country/Region' , 'YearMonth'])
)
monthly_sum_trends
```

	Country/Region	YearMonth	Confirmed	Deaths	Recovered
0	Afghanistan	2020-01	0	0	0
1	Afghanistan	2020-02	1	0	0
2	Afghanistan	2020-03	174	4	5
3	Afghanistan	2020-04	1952	60	255
4	Afghanistan	2020-05	13081	194	1068
...	...	...	...	...	...
3276	Zimbabwe	2021-01	19521	854	14794
3277	Zimbabwe	2021-02	2701	246	6622
3278	Zimbabwe	2021-03	793	60	2020
3279	Zimbabwe	2021-04	1375	44	926
3280	Zimbabwe	2021-05	676	27	966

3281 rows × 5 columns

```
# 7.3 Redo the analysis in the last question for United States, Italy, and Brazil.
us_trends = monthly_sum_trends[monthly_sum_trends['Country/Region'] == 'US']
us_trends
```

	Country/Region	YearMonth	Confirmed	Deaths	Recovered
3043	US	2020-01	7	0	0
3044	US	2020-02	17	1	7
3045	US	2020-03	192276	5369	7017
3046	US	2020-04	888719	60865	146923
3047	US	2020-05	717698	41605	293257
3048	US	2020-06	843456	19767	275873
3049	US	2020-07	1925246	26550	717529
3050	US	2020-08	1459475	29661	746665
3051	US	2020-09	1208533	23399	655863

```
italy_trends = monthly_sum_trends[monthly_sum_trends['Country/Region'] == 'Italy']
italy_trends
```

	Country/Region	YearMonth	Confirmed	Deaths	Recovered
1445	Italy	2020-01	2	0	0
1446	Italy	2020-02	1126	29	47
1447	Italy	2020-03	104664	12399	15683
1448	Italy	2020-04	99671	15539	60216
1449	Italy	2020-05	27534	5448	81562

1457	Italy	2021-01	445866	14357	547437
1458	Italy	2021-02	372233	9183	394651
1459	Italy	2021-03	659634	11647	507846
1460	Italy	2021-04	437754	11461	552531
1461	Italy	2021-05	190402	5195	379511

```
brazil_trends = monthly_sum_trends[monthly_sum_trends['Country/Region'] == 'Brazil']
brazil_trends
```

	Country/Region	YearMonth	Confirmed	Deaths	Recovered
391	Brazil	2020-01	0	0	0
392	Brazil	2020-02	2	0	0
393	Brazil	2020-03	5715	201	127
394	Brazil	2020-04	81470	5805	35808
395	Brazil	2020-05	427662	23308	170620
396	Brazil	2020-06	887192	30280	581763
397	Brazil	2020-07	1260444	32881	1220536
398	Brazil	2020-08	1245787	28906	1259737
399	Brazil	2020-09	902663	22571	1006183

```
# 8.1 For the combined dataset, identify the three countries with the highest average death rates
# (deaths/confirmed cases) throughout 2020. What might this indicate about the pandemic's
# impact in these countries?

df = monthly_sum_trends.copy() # Shallow Copy

df = df.sort_values(['Country/Region','YearMonth'])

# df_2020 = df[df['YearMonth'].between('2020-01','2020-12')]
df_2020 = df[(df['YearMonth'] >= '2020-01') & (df['YearMonth'] <= '2020-12')]
df_2020
```

	Country/Region	YearMonth	Confirmed	Deaths	Recovered
0	Afghanistan	2020-01	0	0	0
1	Afghanistan	2020-02	1	0	0
2	Afghanistan	2020-03	174	4	5
3	Afghanistan	2020-04	1952	60	255
4	Afghanistan	2020-05	13081	194	1068
...	...	...	...	...	...
3271	Zimbabwe	2020-08	3328	135	4217
3272	Zimbabwe	2020-09	1341	26	1082
3273	Zimbabwe	2020-10	529	15	1591
3274	Zimbabwe	2020-11	1583	33	588
3275	Zimbabwe	2020-12	3917	87	2768

2316 rows × 5 columns

```python
# Aggregate Totals [Confirmed , Deaths] over 2020
totals_2020 = (
    df_2020.groupby(('Country/Region'))[['Deaths' , 'Confirmed']]
    .sum()
    .rename(
        columns = {
            'Deaths' : 'Deaths2020',
            'Confirmed' : 'Confirmed2020',
        }
    )
)
totals_2020
```

Country/Region	Deaths2020	Confirmed2020
Afghanistan	2191	51526
Albania	1181	58316
Algeria	3131	99610
Andorra	84	8049
Angola	408	17553
...	...	...
Vietnam	36	1465
West Bank and Gaza	1400	138004
Yemen	610	2100
Zambia	388	20725
Zimbabwe	363	13873

193 rows × 2 columns

```python
totals_2020['DeathRates'] = (totals_2020['Deaths2020'] / totals_2020['Confirmed2020'] * 100).round(2)
totals_2020
```

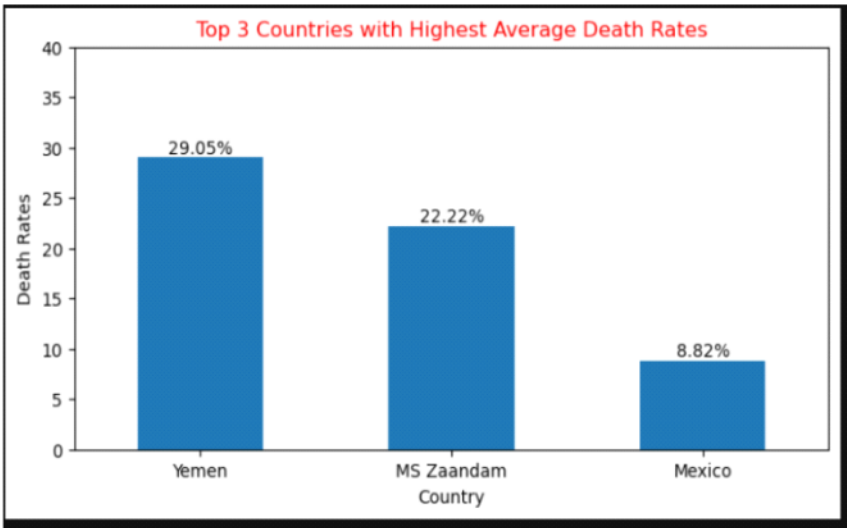
Country/Region	Deaths2020	Confirmed2020	DeathRates
Afghanistan	2191	51526	4.25
Albania	1181	58316	2.03
Algeria	3131	99610	3.14
Andorra	84	8049	1.04
Angola	408	17553	2.32
...	...	...	...
Vietnam	36	1465	2.46
West Bank and Gaza	1400	138004	1.01
Yemen	610	2100	29.05
Zambia	388	20725	1.87
Zimbabwe	363	13873	2.62

193 rows × 3 columns

```python
top_3 = totals_2020.sort_values('DeathRates' , ascending = False).head(3)
top_3
```

Country/Region	Deaths2020	Confirmed2020	DeathRates
Yemen	610	2100	29.05
MS Zaandam	2	9	22.22
Mexico	125807	1426094	8.82

```python
# Visualization [Bar Plot]
ax = top_3['DeathRates'].plot(kind = 'bar' , figsize = (8,4))
ax.set_title('Top 3 Countries with Highest Average Death Rates' , color = 'red')
ax.set_xlabel('Country')
ax.set_ylabel('Death Rates')
ax.bar_label(ax.containers[0] , fmt = '%.2f%%')
plt.xticks(rotation = 0)
plt.ylim(0,40)
plt.show()
```



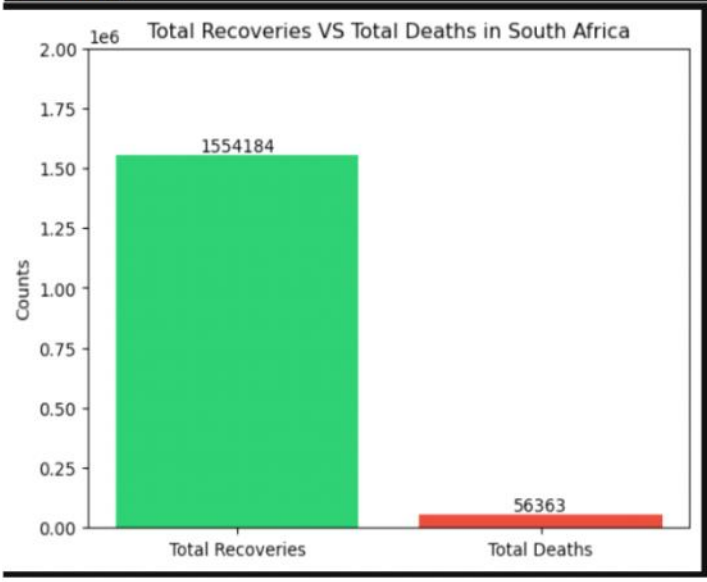

```python
# 8.2 Using the merged dataset, compare the total number of recoveries to the total number of deaths
# in South Africa. What can this tell us about the outcomes of COVID-19 cases in the country?

df = monthly_sum_trends.copy()
total_recoveries = df[df['Country/Region'] == 'South Africa']['Recovered'].sum()
total_recoveries
```

```
1554184
```

```python
total_deaths = df[df['Country/Region'] == 'South Africa']['Deaths'].sum()
total_deaths
```

```
56363
```

```python
# Visualization [Bar Plot]
bars = plt.bar(['Total Recoveries' , 'Total Deaths'] , [total_recoveries , total_deaths],
               color = ['#2ECC71' , '#E74C3C'])
plt.title('Total Recoveries VS Total Deaths in South Africa')
plt.ylabel('Counts')
plt.bar_label(bars, fmt='{:.0f}')
plt.ylim(0,2000000)
plt.show()
```




```python
# 8.3 Analyze the ratio of recoveries to confirmed cases for the United States on a monthly basis
# from March 2020 to May 2021. Which month experienced the highest recovery ratio, and
# what could be the potential reasons?
us_data = monthly_sum_trends.copy() # Shallow Copy
us_data = us_data[us_data['Country/Region'] == 'US'].sort_values('YearMonth').set_index('YearMonth')
us_data
```

YearMonth	Country/Region	Confirmed	Deaths	Recovered
2020-01	US	7	0	0
2020-02	US	17	1	7
2020-03	US	192276	5369	7017
2020-04	US	888719	60865	146923
2020-05	US	717698	41605	293257
2020-06	US	843456	19767	275873
2020-07	US	1925246	26550	717529
2020-08	US	1459475	29661	746665
2020-09	US	1208533	23399	655863
2020-10	US	1930191	24532	771790
2020-11	US	4504713	39271	1537028
2020-12	US	6420021	81142	1151762

```
recovery_ratio_us = (
    us_data['Recovered'] / us_data['Confirmed']
)
recovery_ratio_us = recovery_ratio_us.loc['2020-03' : '2021-05']
recovery_ratio_us
```

```
YearMonth
2020-03    0.036494
2020-04    0.165320
2020-05    0.408608
2020-06    0.327075
2020-07    0.372695
2020-08    0.511598
2020-09    0.542693
2020-10    0.399852
2020-11    0.341204
2020-12    0.179150
2021-01    0.000000
2021-02    0.000000
2021-03    0.000000
2021-04    0.000000
2021-05    0.000000
Freq: M, dtype: float64
```

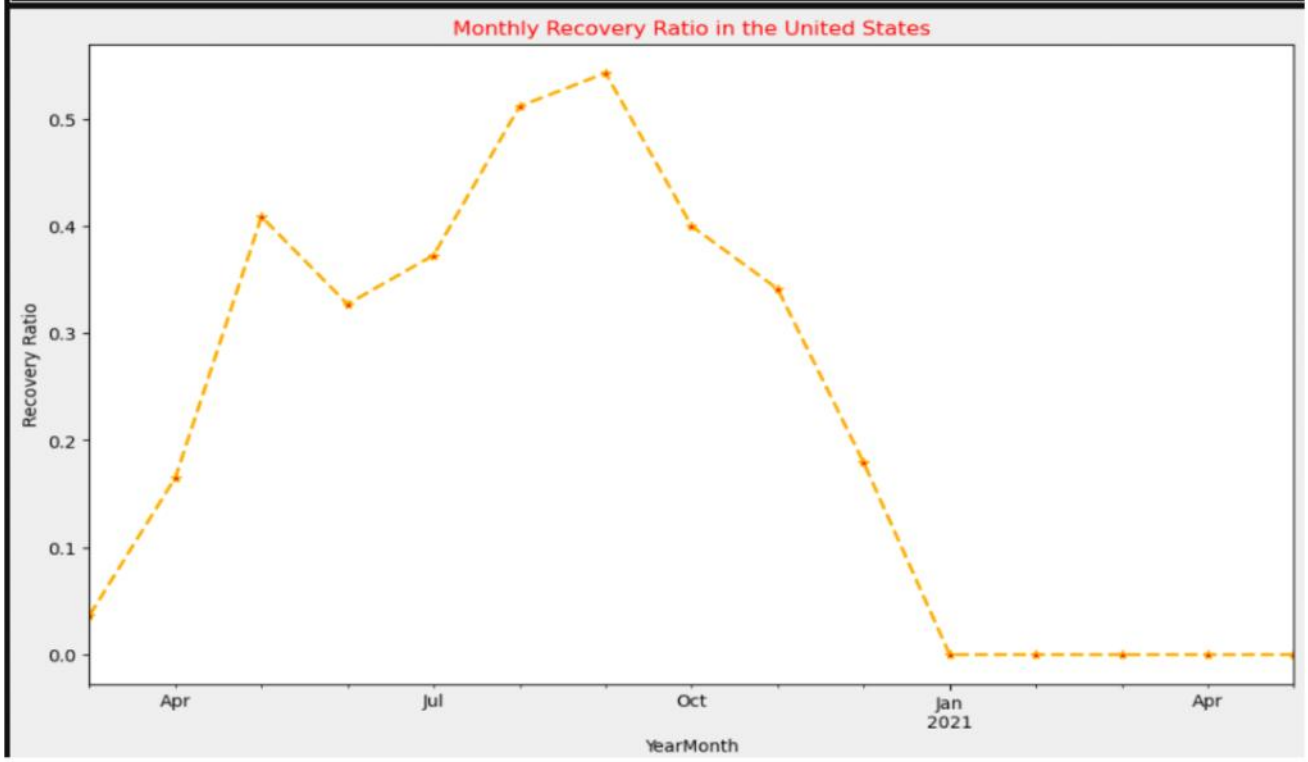
```
recovery_ratio_us.index
```

```
PeriodIndex(['2020-03', '2020-04', '2020-05', '2020-06', '2020-07', '2020-08',
             '2020-09', '2020-10', '2020-11', '2020-12', '2021-01', '2021-02',
             '2021-03', '2021-04', '2021-05'],
            dtype='period[M]', name='YearMonth')
```

```
recovery_ratio_us.values
```

```
array([0.03649441, 0.16531997, 0.4086078 , 0.32707456, 0.37269471,
       0.51159835, 0.5426935 , 0.39985162, 0.34120442, 0.17915033,
       0.        , 0.        , 0.        , 0.        , 0.        ])
```

```
# Plotting the Line Plot [recovery_ratio_us]
plt.figure(figsize = (12,7) , facecolor = '#E8E8E8')
recovery_ratio_us.plot(kind = 'line', marker = '*', markerfacecolor='red', color = 'orange',
                       linewidth = 2, linestyle = '--')

plt.title('Monthly Recovery Ratio in the United States', color = 'red')
plt.xlabel('YearMonth')
plt.ylabel('Recovery Ratio')
plt.show()
```

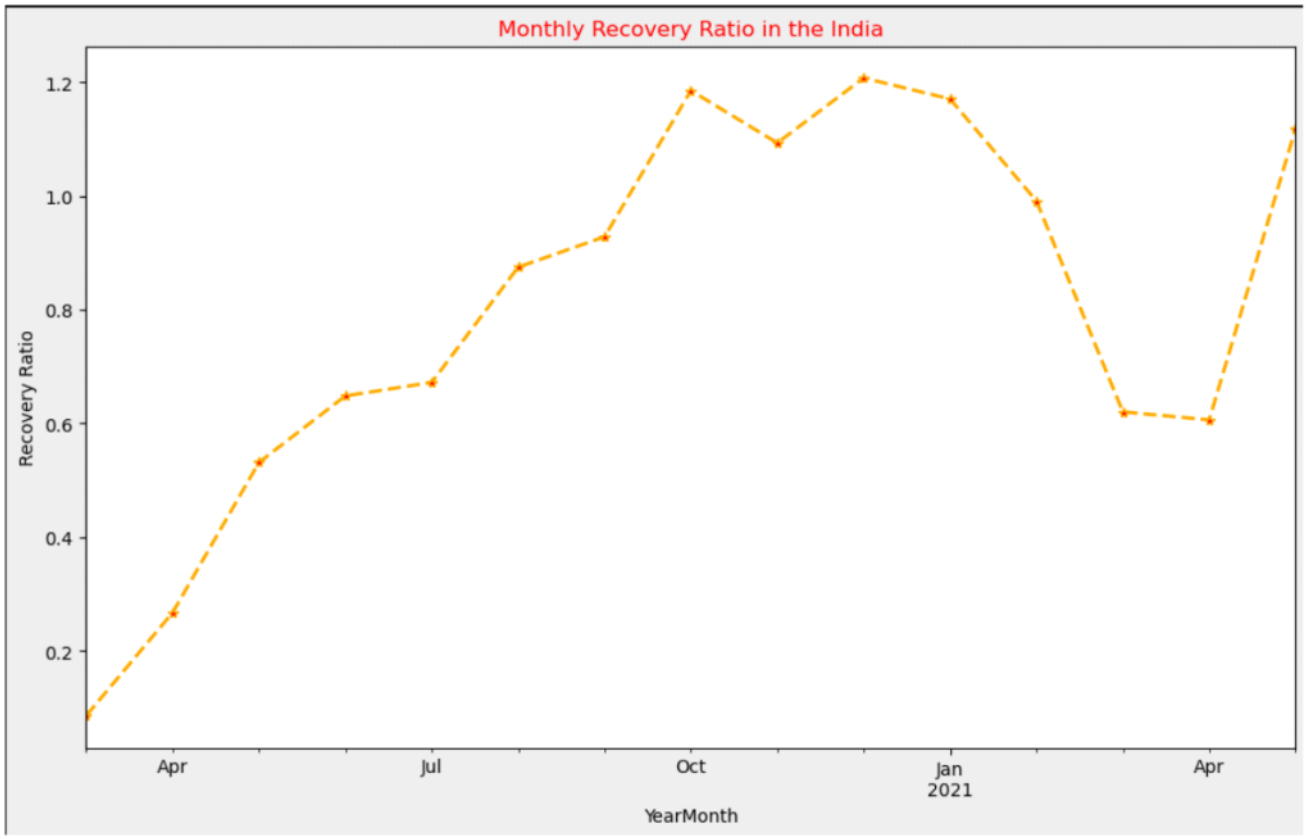


```
india_data = monthly_sum_trends.copy() # Shallow Copy
india_data = india_data[india_data['Country/Region'] == 'India'].sort_values('YearMonth').set_index('YearMonth
india_data
```

	Country/Region	Confirmed	Deaths	Recovered
YearMonth				
2020-01	India	1	0	0
2020-02	India	2	0	3
2020-03	India	1394	36	120
2020-04	India	33466	1119	8945
2020-05	India	155746	4254	82784
2020-06	India	394872	11992	256060
2020-07	India	1110507	19111	746462
2020-08	India	1995178	28777	1745508
2020-09	India	2621418	33390	2433319
2020-10	India	1871498	23433	2218312
2020-11	India	1278727	15510	1398072
2020-12	India	803865	11117	970695
2021-01	India	490936	5654	574703
2021-02	India	354631	2765	351469
2021-03	India	1109424	5770	688231
2021-04	India	6943304	48926	4209723
2021-05	India	8729831	114119	9769914

```
recovery_ratio_india = (
    india_data['Recovered'] / india_data['Confirmed']
)
recovery_ratio_india = recovery_ratio_india.loc['2020-03' : '2021-05']
recovery_ratio_india
```

```
YearMonth
2020-03    0.086083
2020-04    0.267286
2020-05    0.531532
2020-06    0.648463
2020-07    0.672181
2020-08    0.874863
2020-09    0.928245
2020-10    1.185314
2020-11    1.093331
2020-12    1.207535
2021-01    1.170627
2021-02    0.991084
2021-03    0.620350
2021-04    0.606300
2021-05    1.119141
Freq: M, dtype: float64
```

```
# Plotting the Line Plot [recovery_ratio_us]
plt.figure(figsize = (12,7) , facecolor = '#E8E8E8')
```

```
# Plotting the Line Plot [recovery_ratio_us]
plt.figure(figsize = (12,7) , facecolor = '#E8E8E8')
recovery_ratio_india.plot(kind = 'line', marker = '*', markerfacecolor='red', color = 'orange',
                          linewidth = 2, linestyle = '--')

plt.title('Monthly Recovery Ratio in the India', color = 'red')
plt.xlabel('YearMonth')
plt.ylabel('Recovery Ratio')
plt.show()
```